

Question - SetA

1	public class A {
2	public static int temp = 4;
3	public int x = 2, y = 1, sum = 0;
4	public A() {
5	y += temp--;
6	sum = x + y + temp;
7	}
8	public A(int x) {
9	y += x;
10	temp++;
11	}
12	public String methodA(int p, int q) {
13	int z = x + p + (temp++);
14	y = y + q + z;
15	sum = sum + x + y;
16	return x + " " + y + " " + sum;
17	}
18	}
19	public class B extends A {
20	public static int temp = 10;
21	public int x = 6, y = 2;
22	public B() {
23	super(3);
24	y = y + temp;
25	sum = sum + x - y;
26	}
27	public String methodA(int p, int q) {
28	int z = 0;
29	z = x + p + (++temp);
30	y = y + q - z;
31	System.out.println(super.methodA(y, x));
32	sum = sum + x + y + temp;
33	return y + " " + x + " " + sum;
34	}
35	public void methodB(int p) {
36	int y = p + this.y;
37	x = x - y + temp;
38	System.out.println(methodA(x, y));
39	sum = sum + x + y;
40	super.methodA(sum, p);
41	System.out.println(x + " " + y + " " + sum);
42	}
43	}

Illustrate the output of the following statements written in the main method of the driver code. [Answers without workings on the script will be rejected]

```
Driver code:
public class Tester{
    public static void
    main(String []args){
        B b1 = new B();
        b1.methodB(2);
    }
}
```

Solution / Output:

2	24	20
11	2	44
2	14	156

1. Each output yields 1 mark, with a total of 9 marks.
2. A complete table yields 1 mark.
3. If the trace table is not found ⇒ 0 marks.

Question - SetB

1	public class A {
2	public static int temp = 6;
3	public int x = 1, y = 2, sum = 0;
4	public A() {
5	y -= temp++;
6	sum = x - y + temp;
7	}
8	public A(int x) {
9	y += x;
10	temp--;
11	}
12	public String methodA(int m, int n) {
13	int z = x + m + (++temp);
14	y = y + n - z;
15	sum = sum + x + y;
16	return y + " " + x + " " + sum;
17	}
18	}
19	public class B extends A {
20	public static int temp = 8;
21	public int x = 3, y = 5;
22	public B() {
23	super(2);
24	y = y - temp;
25	sum = sum - x + y;
26	}
27	public String methodA(int m, int n) {
28	int z = 0;
29	z = x + m + (temp++);
30	y = y - n + z;
31	System.out.println(super.methodA(x, y));
32	sum = sum + x + y + temp;
33	return x + " " + y + " " + sum;
34	}
35	public void methodB(int m) {
36	int y = m + this.y;
37	x = x + y + temp;
38	System.out.println(methodA(y, x));
39	sum = sum + x + y;
40	super.methodA(sum, m);
41	System.out.println(x + " " + y + " " + sum);
42	}
43	}

Illustrate the output of the following statements written in the main method of the driver code. [Answers without workings on the script will be rejected]

```
Driver code:
public class Tester{
    public static void
    main(String []args){
        B b1 = new B();
        b1.methodB(4);
    }
}
```

Solution / Output:

-9	1	-14
12	6	13
12	1	-12

1. Each output yields 1 mark, with a total of 9 marks.
2. A complete table yields 1 mark.
3. If the trace table is not found ⇒ 0 marks.